Search Beats Imitation: From Offline RL to Neural-Guided Determinized MCTS for the Pokémon Trading Card Game

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Abstract

We study agents for the Pokémon Trading Card Game, a stochastic, imperfect-information game with variable action sets (Kaggle *PTCG AI Battle Challenge*). A first generation combined a relational transformer with imitation, offline RL, self-play PPO and weight ensembling: 63% expert-move accuracy, but at most 650 on the ladder. A second generation uses a compact transformer as prior and value for determinized Monte Carlo tree search on the exact simulator, trained with self-play and expert iteration and evaluated with a fixed gauntlet, confidence intervals and sequential tests. Search raises the gauntlet score from 0.75 to 0.90 and the rating to ≈794. A learned belief over the opponent's hand predicts hidden cards better than counting yet does not improve play: one deep tree beats several shallow determinized ones.

1. Introduction

In the Pokémon TCG two players pilot 60-card decks; hands, deck order and prize cards are hidden, draws and coin flips are random, and the legal move set changes at every decision (play, attach, evolve, attack, choose targets, choose k of n cards). Kaggle provides 5k high-rated replays, a C++ simulator that can step a game forward once the hidden cards are guessed, and a rated ladder; agents get 2 CPUs and 600 s of thinking per game.

We report two generations of agents: the first (§2) asks how far learning alone goes; the second (§3), motivated by its failure modes, asks how much a learned model gains from planning with the real engine. We contribute an account of what transferred from offline metrics to play, a search agent that lifts the rating from 650 to ≈794, controlled ablations (belief, determinization, budget, expert iteration) and an evaluation protocol that agrees with the ladder. Code: `https://github.com/USER/ptcg-agent` (TODO).

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Related work. AlphaGo/AlphaZero combine policy/value networks with PUCT search (Silver et al., 2016; 2018), and expert iteration distils search into the network (Anthony et al., 2017). For imperfect information, perfect-information Monte Carlo samples *determinizations* of the hidden state (Long et al., 2010); information-set MCTS shares one tree across samples (Cowling et al., 2012). We also build on transformers with pointer outputs (Vaswani et al., 2017; Vinyals et al., 2019), IQL (Kostrikov et al., 2022), PPO (Schulman et al., 2017) and WiSE-FT (Wortsman et al., 2022).

2. First generation: learning without planning

Model. *PTCGDualCrossNet* encodes 60 state and 80 option tokens (332-d hand-crafted features with frozen card-text embeddings) and a global vector with a card-tracker belief. A 4-layer encoder (d=256) is followed by 3 cross-attention layers where options attend to the state with biases from 8 symbolic relations (e.g. "evolves from"), a gated self-attention over options and an autoregressive pointer for k-of-n choices.

Training. Behavioural cloning (BC) on 75k decisions from the top ≈650 replays; IQL (expectile τ=0.75, advantage-weighted extraction); league PPO against past snapshots over 5–15 decks; WiSE-FT interpolation of BC, IQL and PPO weights; plus a recurrent variant and an oracle critic. A light search vetoed deck-outs, took lethal attacks and re-ranked close ties (6 ms/move).

Findings. Fixing a one-step observation/action misalignment in the replays raised BC accuracy from 40.9% to 63.3%; IQL reached 63.4%. The extra machinery did not transfer to play: the recurrent model tied the memoryless one (47.5%, 40 games), the oracle critic did not fit values better, PPO saw only ≈165 games, and an agent rated +240 Elo *better* locally (40 games) scored 500 on the ladder

Table 1. System comparison. Expert-move accuracy is measured on each generation's own validation/test split. Gauntlet: score vs the six fixed opponents (95% CI).

System	top-1	gauntlet	ladder
Gen-1 BC / IQL	63.3 / 63.4%	–	–
Gen-1 WiSE + light search	–	–	650
Gen-1 final (search v4)	–	–	500
Gen-2 policy only	70.0%	.75 [.69,.80]	–
Gen-2 MCTS, 4 worlds, 2 s	–	.90 [.85,.94]	737
Gen-2 MCTS, 1 world, 2 s	–	–	768
+ ExIt, 3 s, deck library	–	–	**794**

vs 650 for its predecessor; with 350 games the 650-agent scores only 0.32 locally. Small head-to-head tests between our own agents did not predict the ladder.

3. Second generation: planning with the simulator

Network. We keep entity/option tokens but drop hand-crafted features (option tokens add the embeddings of the cards they act on); a 4-layer pre-norm transformer (d=192, 2.6M parameters) feeds a policy, a count head for k-of-n choices, a value head and a *belief* head predicting which cards are in the opponent's hand, supervised by the opponent's own view stored in the replay. We train on 453k decisions from the top 3,000 replays (split 90/5/5 by game). The value head overfits the 3k game outcomes, so it is fine-tuned on 8k self-play games with the policy anchored on replays.

Search. At each decision we sample K determinizations (own deck/prizes uniform; opponent decklist the most consistent of 120 decklists mined from replays and ladder games; opponent hand uniform or from the belief head) and run PUCT with network priors and value leaves on the simulator; forced moves are played by the policy and root visits are summed over worlds. A time manager spends $\min(c, (T_{\text{left}} - r)/q)$ seconds per decision. Expert iteration (ExIt) fine-tunes the network on visit distributions of MCTS self-play.

Protocol. Offline metrics on held-out games; online, a fixed gauntlet of six opponents (two first-generation agents and our policy piloting four meta decks) with a simulated time budget, 95% Wilson intervals, paired head-to-head ablations and an SPRT (H_0: 0, H_1: +30 Elo).

4. Results

Offline and deck. The Gen-2 policy predicts 70.0% of expert moves (uniform: 21.7%; two seeds within 0.3 points); the belief head beats counting even with the true decklist, and self-play makes the value better than a constant (Tab. 2). The deck dominating top replays (Dragapult) scores 0.49 with our policy, Kangaskhan/Crustle 0.73 (240 games each).

Table 2. Gen-2 offline results (held-out games) and online ablations (score of A vs B, 95% CI).

Policy top-1, main phase ≥5 options (uniform)	60.7% (11.6%)
Value Brier: replay-only / +self-play (const.)	.254 / .226 (.249)
Belief AUROC: learned / count w/ true list	.752 / .711
MCTS-64 vs own policy (200 games)	.59 [.52, .66]
learned vs count-based hand sampling (400)	.48 [.43, .53]
4 worlds vs 1 world, equal sims (300)	.39 [.34, .45]
ExIt-2 + MCTS vs base + MCTS (300)	.57 [.51, .62]
ExIt-3 + MCTS vs ExIt-2 + MCTS (300)	.50 [.45, .56]
searching k-of-n choices vs policy (400)	.48 [.43, .53]

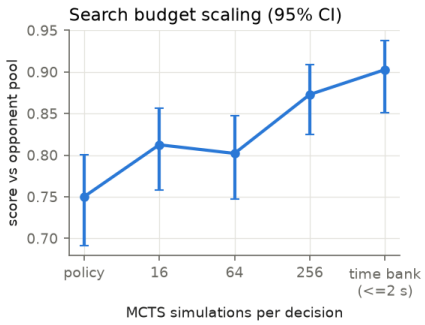


Figure 1. Gauntlet score vs search budget (Crustle deck, 180–240 games per point, 95% CI); "time bank" is the deployed agent.

Search. Search is the main source of strength (Fig. 1): 0.75 for the raw policy, 0.87 at 256 simulations and 0.90 for the deployed agent, reproduced on a second seed. Belief-sampled hands do not help, and four determinizations lose to one deeper tree (-75 Elo): at these depths the exact hand rarely matters. ExIt with 64-simulation targets leaves the raw policy unchanged but strengthens search (+49 Elo); a third round and searching k-of-n choices did not help.

Ladder. Gen-2 agents rate 737–794 against 500–650 for Gen-1 (Tab. 1), and the local ranking (1 world $>$ 4 worlds, +46 Elo over 400 games) matches the ladder.

5. Discussion and conclusions

Learning alone plateaued: richer architectures and offline/online RL improved imitation metrics but not play, while a smaller network used as a prior for search on the true game engine gained ≈150 rating points. Belief modelling helps prediction, not play, at our depth. Limitations: four of six gauntlet opponents share our network, and 800+-rated opponents remain hard (0.36–0.50).

6. Statement on the use of AI

I used an AI assistant (Claude, Anthropic) extensively throughout the project, as a pair programmer and research assistant, in both generations. It helped write and debug code (data pipelines, models, training, search, evaluation), diagnose failure modes, propose experiments, launch and monitor runs on my machine, analyse results, produce plots and draft this report. I chose the directions of the project, reviewed code and results, ran the Kaggle submissions and checked that every number reported comes from logged experiments (`RESULTS.md`, `PROJECT_JOURNAL.md`); I take full responsibility for the content and can explain all of it.

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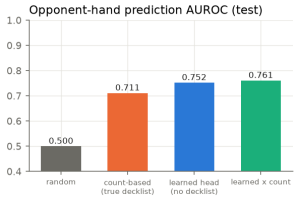
Appendix A: first-generation details

Table 3. Gen-1 local Bradley–Terry ladder (mirror deck, 20 games per pair; Elo relative to the mean). Note the small samples; the top-rated agents later scored lower on the ladder than `wise`.

Agent	BT Elo	mirror win rate
self-play PPO v3	+180.5	70.6%
search v4 (WiSE v3)	+120.7	62.5%
WiSE v3 (IQL + PPO + IQL-rel.)	+120.7	62.5%
IQL v3	+116.1	61.9%
IQL relational	+111.6	61.3%
BC	+107.1	60.6%
WiSE v1 (ladder 650)	+44.0	51.9%
heuristic	−296.1	15.6%
first submission (ladder 250–400)	−504.6	3.1%

Other Gen-1 components: a championship-deck tournament over decks reconstructed from top replays (16 games per pair, which selected Hydrapple/Ogerpon); a card tracker with archetype inference; rules for multi-select counts (e.g. attaching 2–5 energies); and forensic analysis of ladder losses, which found the replay misalignment, a module-name collision that crashed a submission, belief-state carry-over between games of a match, and deck-search choices blind to evolution lines. The recurrent policy tied the memoryless one (47.5%, CI [32.9, 62.5], 40 games); the asymmetric (oracle) critic did not fit values better than the observable critic.

Appendix B: additional second-generation results



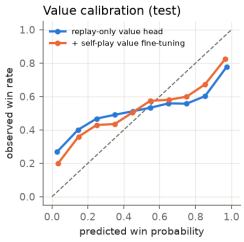


Figure 2. Left: opponent-hand prediction AUROC on held-out games. Right: value calibration before and after self-play fine-tuning.

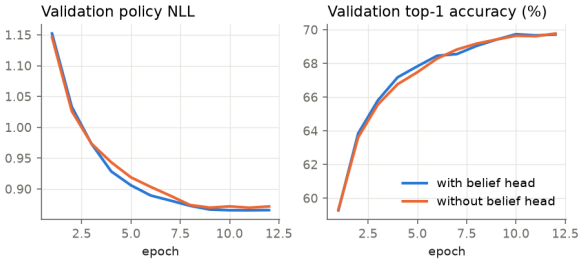


Figure 3. Validation curves with and without the auxiliary belief head (no effect on the policy).

Table 4. Further ablations (Crustle mirror; score of A vs B, 95% CI).

learned vs count-based hands, 256 sims (208)	.52 [.45, .58]
deployed: 1 vs 4 worlds (400)	.57 [.52, .61]
$c_{\text{puct}} = 0.7$ vs 1.25 (240)	.51 [.45, .58]
$c_{\text{puct}} = 2.5$ vs 1.25 (240)	.43 [.37, .49]
3 s cap + 120-deck library vs 2 s, 8 decks (160)	.54 [.46, .61]

Table 5. Deck selection: our policy piloting each deck against the gauntlet (240 games each).

Mega Kangaskhan ex / Crustle	.73 [.67, .78]
Teal Mask Ogerpon ex	.65 [.59, .71]
Hydrapple ex	.64 [.58, .70]
Marnie's Grimmsnarl ex	.59 [.53, .65]
Dragapult ex	.49 [.43, .56]
Lillie's Clefairy ex	.28 [.23, .34]